

Quantum Image Hackathon

Rule Book

Exploring the use of quantum computing in
quantum image analysis and quantum image generation

Quantum AlgoLab
iQuCodeFest 2026
June 1, 2026

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1 Context

Images are one of the most important and information-rich data modalities in modern computing. They are used in medicine, remote sensing, industrial inspection, scientific visualization, media, and generative design. At the same time, image problems are often high-dimensional, noisy, and computationally demanding. This hackathon invites participants to explore whether quantum computing can contribute meaningfully to image workflows through two tracks:

- **Quantum image analysis** – detect, classify, segment, reconstruct, compare, or enhance images using hybrid quantum-classical methods.
- **Quantum image generation** – create or synthesize images using quantum or hybrid quantum-classical methods, including probabilistic, variational, or generative approaches.

The goal is not to force quantum methods into every pipeline. Instead, participants are expected to identify a problem where quantum ideas are pertinent, justify the choice clearly, and build a concrete proof of concept.

The best submissions will combine technical rigor, implementability, and creativity. A strong project will explain *why* the quantum component is needed, *where* it fits in the pipeline, and *how* it is validated.

2 Rules

All teams should follow the workflow below, regardless of the track selected.

Step 1: Choose a track Select one track or propose a solution that spans both tracks.

- **Track A - Quantum image analysis**
- **Track B - Quantum image generation**

A team may submit only one track, one project that addresses both tracks, or two complementary subprojects, one per track, provided the link between them is clearly explained.

Participants are encouraged to choose a scope that can be implemented within the hackathon timeframe. A narrower project that works end-to-end is preferred over a broad concept that remains unimplemented.

Step 2: Define the image problem precisely Whether track A or B is chosen, a clear problem statement is essential. Examples include image classification, anomaly detection, denoising, segmentation, retrieval, super-resolution, image completion, texture synthesis, or conditional generation.

Describe the image task in clear terms. The solution must specify:

- the input format (for example: grayscale images, RGB images, patches, latent codes, segmented masks, or feature vectors),
- the target output,
- the expected use case,
- the baseline classical approach,
- the metric or criteria used to evaluate success.

Step 3: Identify the quantum contribution The quantum part must be more than decoration. Teams must explain the specific role played by quantum computation. Here are some examples of what exists in the literature, with various levels of pertinence :

- quantum feature maps for image descriptors,
- variational quantum circuits for classification or generative modelling,
- quantum kernels for similarity and retrieval,
- quantum sampling for image generation,
- quantum optimization for model selection or reconstruction.

Teams should justify the choice of representation and algorithm. You should ask yourselves why is this quantum method useful for the chosen image task?

Step 4: Design a practical pipeline The project must be implementable. Teams should design a pipeline that can be executed or simulated. A typical pipeline will probably include data loading and preprocessing, measurement and decoding and post-processing or evaluation. Hybrid architectures are encouraged when they improve realism.

For example a classical encoder followed by a quantum latent processor, or a quantum model used only for a specific subproblem such as similarity scoring, sampling, or decision refinement.

Step 5: Build a proof of concept Each submission must include a working proof of concept. The proof of concept should demonstrate the core idea of the project, even if the full-scale system is not done/possible.

The proof of concept should be well documented. It can use a simulator, a noise model, or even a limited hardware demonstration.

2.1 Track-specific requirements

2.1.1 Track A: Quantum image analysis

Projects in this track should address an analysis task on image data. Examples of valid tasks include classification, segmentation, object localization, anomaly detection, denoising, compression, image similarity, feature extraction...

A valid solution can use a quantum model at any stage of the pipeline, provided the role is justified. Some options include quantum kernels, variational classifiers, quantum-enhanced embeddings, or quantum-assisted clustering.

2.1.2 Track B: Quantum Image Generation

Projects in this track should generate or synthesize visual content. Suitable tasks include image generation, image completion, texture synthesis, and related forms of visual content creation.

The generative system may be fully quantum, hybrid quantum-classical, or quantum-assisted. Strong submissions should clearly identify the role of the quantum component: what part of the image-generation pipeline it performs, what is produced by it, and how the quality of the generated output is evaluated.

A project may still be considered strong even if the quantum component is relatively small, provided that it is technically well motivated and implemented soundly. The image-generation task should also be justified and contextualized: submissions should explain why this particular generative approach is being developed and what purpose it serves.

Image generation as an artistic or exploratory practice is also acceptable. For example, a project focused on quantum-assisted fractal generation may be a valid application if the artistic goal is stated and the generative process is well explained.

3 Submission requirements

Each team must submit a complete package containing the following elements.

3.1 Project presentation

The presentation should explain the project from end to end. Teams may provide visual examples or even a short demo video. The presentation should include:

- the selected track or tracks,
- the image problem being solved,
- the classical baseline,
- the quantum approach/hybrid architecture,
- the proof of concept,
- the evaluation method,
- the limitations and/or future directions.

3.2 Working proof of concept

A working prototype is required. The prototype should be sufficient to demonstrate the main idea and should be reproducible by the judges.

Implementation format is a notebook describing your solution paired with source code. The submission should make it easy to run the project, inspect the solution, and observe the output.

The notebook outputs must be pushed to your repository so that a look at your document should provide a good overview and understanding of your solution. You should provide a concise run guide. The instructions must give the required dependencies, how to run the prototype and how to reproduce the reported resultss.

4 Judging criteria

Submissions will be evaluated according to the criteria below.

Criterion	Description
Creativity	How original and insightful the image problem formulation and quantum integration are.
Feasibility	Whether the proposal is technically plausible and implementable within realistic constraints.
Quantum relevance	Whether the quantum contribution is justified, non-trivial, and appropriate to the task.
Technical implementation	Whether the proof of concept functions correctly and demonstrates the core mechanism.
Pipeline clarity	Whether the entire pipeline is understandable from input to output, including the role of each component.
Presentation quality	Whether the team communicates the idea clearly and effectively.

A strong project does not need to be the most advanced quantum system. It should, however, be a well thought-out solution in which the quantum portion earns its place.